

Advanced Calculus I and II Tailored For Two 10-Week Terms

Scott F. Beaver

Western Oregon University

Contents

Preface	iii
To the Instructor	iv
1 Preliminaries	1
1.1 Notation	1
1.2 Fundamental Definitions and Axioms	1
2 Sequences of Real Numbers	6
3 Series of Real Numbers	10
4 Functions	13
5 Sequences and Series of Functions	16
6 Differentiation	19
7 Integration	22

Preface

These notes were developed for a two-term (two-quarter) sequence (20 weeks total) in Advanced Calculus I & II which I've taught at Western Oregon University (WOU). The course sequence is a requirement for math majors and is typically taken by seniors and occasionally by juniors; the class size is usually about 8-12 students. It is a prerequisite for Complex Analysis, and its scope is restricted to real sequences, series, and functions of the real numbers. The treatment of Advanced Calculus in these notes is fundamentally based on sequences and does not rely on topological notions. The notes have been built from material culled from a variety of sources, including *Elementary Analysis: The Theory of Calculus* by Kenneth A. Ross, notes I took as an undergraduate while enrolled in the modified Moore method course in Real Analysis at Auburn University at Montgomery taught by Jimmy Nanney in 1994, *Understanding Analysis* by Stephen Abbott, and other sources.

Although I have not taught from these notes under a semester schedule, it is my opinion that they are of insufficient length for use in a two-semester course (unless the class size is small enough), and implementation of the notes in such a system would require augmentation by e.g., treatments of countability, metric-space topology, improper integrals, Taylor series, etc. Please feel free to contact me for my opinion on which of these topics would dovetail most effectively with your particular semester-long courses if you'd like.

Acknowledgments

I extend my sincerest gratitude to Harry Lucas, Jr., The Educational Advancement Foundation, and in particular The Legacy of R.L. Moore Project for their advocacy of inquiry-based learning and for funding and organizing the annual Legacy of R.L. Moore conferences. I especially thank Meghann Barger and Darrel Palke for their hours of analysis of and thoughtful input to these notes. I am grateful to Jimmy Nanney of Auburn University at Montgomery for introducing me to the Modified Moore method, and the Mathematics Department at Western Oregon University for being very supportive.

To the Instructor

Prerequisites

The prerequisites for Advanced Calculus at WOU are a four-term sequence in calculus including one term of elementary sequences and series and one term of multivariate calculus, and Introduction to Proofs. In particular, a working knowledge of mathematical induction is assumed for a small number of problems.

Theorems and Problems

The material in these notes on which students can produce assessable work is divided into theorems and problems. Several of the theorems would be more appropriately termed lemmas or corollaries but are not labeled as such to discourage students from targeting problems based on any perception of difficulty. The problems are generally designed to provide concept-image support and are useful but not essential to the course material (although it could be argued, for example, that Exercise 201 leads to an easier, if not more intuitive, proof of The Chain Rule, Theorem 210). It is also worth noting that these notes are hardly exhaustive in the sense that there are many problems, such as the n -term triangle inequality, which are absent but sufficiently taxing and relevant as to provide good exam material. Another device implicit in the notes is that lemmas which do not appear explicitly in the notes are needed in proofs of some of the theorems, so that students must formulate and prove those lemmas (for example the result that if $ab = 0$ then either $a = 0$ or $b = 0$ does not appear explicitly but is indispensable). Compelling students to formulate and then prove intermediate results helps provide students with a good sense of what mathematics is. We name each such intermediate lemma after the student who proves it, giving the prover a measure of pride at being cited later during the course.

Some Suggestions for Adapting These Notes to Your Course

Should an instructor using these notes find that fewer than ten students have enrolled in the sequence, consideration may be given to treating many of the theorems in Section 1.2 as axioms and/or minimizing the role of the problems in the course, especially if one has the goal of reaching the Fundamental Theorem of Calculus.

There is enough variation in the role of this course in the mathematics curriculum across different universities and colleges, between instructors, and between classes of students that a few remarks are in order regarding other potential modification to these notes. Af-

ter some of the more difficult theorems, some hints are given; these can be deleted or expanded. Along those lines, the lemmas disguised as theorems can be deleted for more advanced groups of students who might reasonably be expected to discover the necessity of these lemmas. Alternatively, the opposite approach may be appropriate. Some theorems in these notes could be broken down further into one or more lemmas and a main result. Another way to shrink the content somewhat is to remove that material regarding $\limsup$ and $\liminf$, weaken the Root and Ratio Tests a bit, and expect students to work towards the more standard proof of the Bolzano-Weierstrass Theorem.

These notes are essentially devoid of discussion points, save for a few brief remarks scattered here and there. A wealth of explanatory material has been written in texts and on the internet about the broad ideas covered herein, and students in any upper-division math course - even a modified Moore method course - should be given the opportunity to gain a panoramic view of the forest, not just the beautiful trees. My personal preference is that students acquire this view in a manner consistent with the Moore approach. Ideally, these broad questions should arise naturally as the students spend significant time deeply engaged in the material. In practice, it is necessary that I wait for appropriate opportunities and ask as few leading questions as is necessary to get students to understand that there is a deep or broad connection within reach, and if we don't get it right away, I let them sleep on it and we tie it all up the next day. Sometimes I pose these broader questions as extra-credit essay-type questions on quizzes, sometimes verbally in class after an important theorem has been proved, and often in emails to the class. I particularly like the latter delivery since it usually allows them to first consider the question in solitude.

What Suffices for Justification of Statements in Solutions

It is crucial to develop a culture of justifying everything precisely in the first few weeks of Advanced Calculus I (specifically regarding the results in Section 1.2). This leads to tedious proofs, but also to an awareness that we can take nothing except axioms and proven results for granted. Once the students show that they can perform capably under these circumstances, I slowly free them from the encumbrance of citing basic results. This enables presentation of increasingly complicated proofs in a timely manner. Relaxing the justification threshold must be done unambiguously; as the requirements are loosened we keep a running list of any axioms, definitions, problems, or theorems which no longer require citation. Further, since efficiency does play an important role in both the courses and the mathematical maturation of students, I actively discourage students from citing axioms, definitions, problems, or theorems which are on the "no-cite" list.

Student Assessment

The device of weekly quizzes has proven particularly useful for ensuring that students are sufficiently conversant with definitions and theorem statements. Typically, the content of the quizzes is limited to definition and theorem statements, true-false questions, calculating the $\limsup$ or $\liminf$ of a particular sequence, applying the Ratio Test, or writing a simpler proof of an earlier result given new proven theorems (*e.g.*, using the Cauchy Condensation Test to provide an alternate proof of the divergence of the harmonic series, given that this result had already been demonstrated by the standard technique of considering

partial sums of the form s_{2^n} ; the Integral Test method is certainly off-limits at this point in the course since integrals have not even been defined). The first quiz is during the second day of class and covers the material in the syllabus, in particular the rules of order for the class.

Exams are given, but constitute a small proportion of students' numerical course grades. Presented proofs or problems constitute 60% of the final course points. Each one-directional proof or exercise solution is assigned a score between 5 and 10, depending on difficulty and the cleverness of the presented proof or solution; a bi-directional proof in which both directions are very difficult can then receive a maximum score of 20 points. Students submit their completed written work immediately prior to its presentation and must present their proof/solution without reference to their paper; should a student find him/herself stuck and need to refer to their paper, a penalty of 20% of the total points for the problem is assessed for each such reference, up to one minute each. This policy appears to provide greater ownership of proofs and solutions. During the first (second) term, it is necessary but not sufficient to present four (five) complete solutions, and to pass all quizzes with 100% (students may retake quizzes until this goal is reached), for a passing grade.

Rules of Order

The following is an excerpt from my most recent syllabus for the course.¹ Please note that these rules are only one particular manner in which this course could be run. More generally, Modified Moore method courses can be structured in a broad variety of ways and instructors new to the method should consult several sets of course notes or instructors experienced in this method before choosing or developing a style for their courses. Naturally, we want to be fair and remain cognisant of the objective of producing independent learners, but we shouldn't be afraid of taking creative risks with the set of rules, nor of seeking student input.

Resources: *Your mind and math skills. You may not work together, refer to any texts, or use the internet for the purpose of developing theorem proofs or problem solutions. Do not talk to each other about any problems which are still unsolved in class. Do not write any attempts, successful or otherwise, at a solution to a problem on any vertical surface anywhere outside of class (except possibly alone in your own home). You are on the honor system, as far as you know.*

The main goal of this course is to help you develop the ability to synthesize facts about real numbers, independently. If you use a proof you have found in a book or somewhere else, you likely won't be able to answer the questions asked of you by your colleagues and/or me, unless you have spent about as much time internalizing the pirated proof as you would have spent generating it on your own. If you are good enough to manage to get away with using a looked-up proof or solution, then you must be pretty smart; do this

¹Many of the guidelines in this section are taken from Robert Kauffman's syllabus published in the Journal of Discovery Learning in Mathematics.

and you will have managed to successfully cheat yourself out of the intelligence you might have otherwise gained. *This course can give you the confidence to become an independent thinker and defend yourself intellectually, but only if you follow the rules.* Just to be clear: if I think I've caught you cheating on a proof/solution or on an exam you'll likely be referred to Judicial Affairs. Don't be foolish enough to risk missing graduation rather than seeing me for help. In general, don't violate anything in the Standards of Conduct in the WOU Catalog.

PLEASE TURN OFF YOUR CELL PHONE BEFORE YOU ENTER THE CLASS-ROOM.

Rules:

Psychological:

- Check your ego or diffidence, whichever applies, at the door.
- Don't be too quick to think you are wrong, and don't be too stubborn about being right.

Procedural:

- I will call for volunteers for each problem, and choose two presenters from among those who have the fewest total presentation points, except for the exception mentioned below. Presenters write their proofs in succession; the second presenter must leave the room while the first presenter addresses the class.
- Any student who has not presented during the current class or the preceding one will be chosen ahead of any student who has - this is the exception mentioned above.
- Please submit to me a copy of your proposed proof, or the original, when you begin your presentation.
- There is no concrete penalty for an incorrect solution or failing to answer a question correctly, except that you may not be able to present on that day or the next because of the previous rules.
- If you have a solution to present, and time does not permit that you present it during the current class period, you may, but are not guaranteed to be able to present it during the next class meeting, subject to the rules above.
- To help prove a theorem or solve a problem, you may use results which precede your theorem/problem in the notes even if that result is not yet proven. You may not use results which in the notes succeed the result you are trying to prove.

Compensational:

- The points for proofs/examples/problems will be at least 5 points for each problem (I consider each direction of a bidirectional proof one problem) and will vary based on my perception of the difficulty of proving the problem, *but will not be revealed until after the solution is complete and reviewed by me after class.*
- Each time you consult your paper for the solution, while you are at the board or projector, will result in the loss of one-fifth of the relevant points available.
- You will have one minute to answer each question about your proof. If you cannot answer within one minute or answer incorrectly, you will be "on probation" for that problem/solution, and if you cannot answer another question, I'll ask you to sit down. I may at this point give you no points for your efforts, but in some cases I will award you some

fraction, at least one-half, of the available points, if you have shown sufficient work (more than half) which would lead to a complete solution but is incomplete. In this case, I'll commit to you the rights to the problem's remaining points for 1-2 class periods (depending on my perception of the complexity of the remainder of the solution). If you cannot complete the solution within the given deadline, I will either complete the solution myself for the class (in the case of a relatively trivial remaining part) and award no further points for that problem, or open the remaining points to the rest of the class.

Suggestions:

- It is important that you *internalize* your proof/solution beforehand, and in complete detail, anticipating any difficulties beforehand. During office hours I may give you *very* limited help to do this, but I will always offer as much help as possible for understanding proofs already presented in class and the statements of theorems and definitions.
- Though you may not work together on original solutions or proofs, you are encouraged to work together, if you so desire, for the purpose of studying for quizzes or exams. This is not meant to be taken as license to mooch off note-card makers the day before the exam or quiz! If you want to work together with someone under these circumstances, bring something to the table yourself.

I cannot stress the items in the following section enough. This course can be competitive but it is incumbent on each of you to be respectful of each other! If two boxers who slug it out can congratulate each other in mutual respect afterwards, so can you each compete from positions of respect and appreciation rather than malice or bitterness.

Things We (or You) Don't Do

Disrespect the material:

- ***Let arguments which we don't understand proceed.***
- Use undefined concepts.

Disrespect each other:

- Use the words "obvious" or "stupid."
- Try to intimidate people who question our arguments.
- Get mad at others (but if we do, we get over it quickly).
- Get mad at ourselves for making a mistake (but if we do, we get over it quickly).

Misunderstand your role:

- Test a presenter's understanding of their own proof by asking questions whose answers are already known to you - that is my job alone.
- Try to catch cheaters - that is also my job alone.
- Take notes during presentations. The proofs or problem solutions will be posted within a few days to the course notes; available on the K Drive. Those are your notes. Once received, you may annotate them as you see fit. In particular, put questions on them!

Things We Do

- Learn from our mistakes.
 - Permit proofs to be written up on the board or the document camera without interruption, *at the discretion of the presenter*. If this is the case, those not presenting are to follow what is being written, make notes of any questions they have, and ask them at the conclusion of the *presentation* (not the transcription) of the proof. *I may ask students who are not presenting to explain steps they see on the board or document camera if I have a suspicion that they do not understand something but have failed to ask a question about it.* I possess some skill in detecting confusion (mostly due to personal familiarity with it).
 - Let people answer the questions asked of them.
 - Let people know when we think that they have done something clever.
 - When using an earlier result, cite it specifically by problem or theorem number (or name).
 - Write our arguments in a clear, step-by-step manner, in either English or mathematical sentences, so that our colleagues can analyze them more easily.
-

Privacy

As the courses progress, we maintain two sets of course notes. The first, available on the course webpage and accessible to anyone with internet access, consists of the list of axioms, definitions, theorems, and problems below. The second contains the notes below and the student solutions, credited by name. To maintain privacy of students' work, that document is maintained on the WOU network and is available only to students currently enrolled in the course.

Chapter 1

Preliminaries

“I hear and I forget. I see and I remember. I do and I understand.” - Confucius

1.1 Notation

- $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ the natural numbers, integers, and rational numbers, respectively
- $a \in A$ “ a is a member of the set A ”
- $A \subseteq B$ “ A is a subset of B ”
- $A \cup B$ the union of A and B
- $A \cap B$ the intersection of A and B
- $A \setminus B$ all elements in the set A but not in the set B
- A^c the complement of A with respect to some universal set
- $\emptyset$ the empty set; set with no elements
- $A := B$ “ A is defined as B ”

1.2 Fundamental Definitions and Axioms

Definition 1. (*Axiom*) A proposition assumed true without proof.

Definition 2. (*Theorem*) A proposition proved from other propositions whose truth is known or assumed.

Definition 3. (*The Real Number System*) A set, denoted $\mathbb{R}$, whose elements are called real numbers and on which we define two operations addition (+) and multiplication ($\cdot$) and an order $\leq$, satisfying Axioms 5-12, 15-19, and Axiom 31.

Remark 4. We will usually, but not always, write xy for $x \cdot y$.

Axiom 5. (*Closure under addition*) If $x, y \in \mathbb{R}$, then $x + y \in \mathbb{R}$.

Axiom 6. (Closure under multiplication) If $x, y \in \mathbb{R}$, then $xy \in \mathbb{R}$.

Axiom 7. (Commutativity) If $x, y \in \mathbb{R}$, then $x + y = y + x$ and $xy = yx$.

Axiom 8. (Associativity) If $x, y, z \in \mathbb{R}$, then $x + (y + z) = (x + y) + z$ and $x(yz) = (xy)z$.

Axiom 9. (Existences of Identities) There exists an element $0 \in \mathbb{R}$ such that for all $x \in \mathbb{R}$, one has $x + 0 = x$; also there exists $1 \in \mathbb{R}$ such that for all $x \in \mathbb{R}$ one has $x \cdot 1 = x$.

Remark 10. For any $x \in \mathbb{R}$, we take $-x$ to mean $(-1)x$.

Axiom 11. (Existence of Inverses) For all $x \in \mathbb{R}$ there exists $(-x) \in \mathbb{R}$ such that $x + (-x) = 0$; for all $x \in \mathbb{R} \setminus \{0\}$ there exists $x^{-1} \in \mathbb{R}$ such that $x \cdot x^{-1} = 1$.

Axiom 12. (Distributive laws) If $x, y, z \in \mathbb{R}$, then $x(y + z) = xy + xz$ and $(y + z)x = yx + zx$.

Definition 13. We define subtraction as the addition of the additive inverse: If $x, y \in \mathbb{R}$, then $y - x := y + (-x)$. We define division by $x \neq 0$ as multiplication by the multiplicative inverse: If $x, y \in \mathbb{R}$, $x \neq 0$, then $\frac{y}{x} := yx^{-1}$.

Definition 14. The statement $x < y$ is left rigorously undefined, but we will take it to mean exactly what you have been led to believe. The statement $x \leq y$ means that either $x = y$ or $x < y$; also $y \geq x$ has the same meaning as $x \leq y$ and $y > x$ has the same meaning as $x < y$.

Axiom 15. If $x, y \in \mathbb{R}$, then either $x \leq y$ or $y \leq x$.

Axiom 16. If $x, y \in \mathbb{R}$ with both $x \leq y$ and $y \leq x$, then $x = y$.

Axiom 17. If $x \leq y$ and $y \leq z$, then $x \leq z$.

Axiom 18. If $x \leq y$, then $x + z \leq y + z$.

Axiom 19. If $x \leq y$ and $z \geq 0$, then $xz \leq yz$.

Theorem 20. If $x \leq y$, then $-y \leq -x$.

Theorem 21. If $x \leq y$, and $z \leq 0$, then $yz \leq xz$.

Theorem 22. If $x, y \geq 0$, then $xy \geq 0$.

Theorem 23. For all $x \in \mathbb{R}$, $x^2 \geq 0$.

Problem 24. $0 < 1$.

Theorem 25. If $x > 0$, then $x^{-1} > 0$.

Theorem 26. If $0 < x < y$, then $0 < y^{-1} < x^{-1}$.

Remark 27. Each of the above from Axiom 5 to Theorem 26 is true of $\mathbb{Q}$ as well. What makes $\mathbb{R}$ different? Read on.

For the next three definitions/problems, let $S \subseteq \mathbb{R}$.

Definition 28. (*Upper bound*) If there exists $M \in \mathbb{R}$ such that for all $x \in S$, one has $x \leq M$ then M is an upper bound for S .

Definition 29. (*Least upper bound, supremum*) Suppose S is bounded above (has an upper bound). If there exists $s \in \mathbb{R}$ such that s is an upper bound for S and for all upper bounds M of S , one has $s \leq M$, then we call s the least upper bound or supremum of S , denoted $\sup S$.

Problem 30. Define the analogous concepts of lower bound and greatest lower bound (infimum).

Axiom 31. (*Completeness Axiom*) Let $S \subseteq \mathbb{R}$ be nonempty and bounded above. Then $\sup S \in \mathbb{R}$.

Remark 32. The Completeness Axiom is the most important of all the axioms in the sense that it distinguishes $\mathbb{R}$ from $\mathbb{Q}$. Though the Greeks were aware of irrational numbers, the absence of the Completeness Axiom from Greek mathematics may be a significant part of the reason that they did not discover calculus.

Problem 33. Show that the Completeness Axiom does not hold when $\mathbb{R}$ is replaced by $\mathbb{Q}$.

Definition 34. Let $S \subset \mathbb{R}$. If there exists $M \in S$ such that for all $s \in S$, $s \leq M$, then we say that M is the maximum of the set S , and write $M = \max S$. If there exists $m \in S$ such that for all $s \in S$, $s \geq m$, then we say that m is the minimum of the set S , and write $m = \min S$.

Remark 35. In some texts the Archimedean Property (defined in the next theorem) is given as an axiom. It is presented here as a theorem since in using the Completeness Axiom in its proof, we will establish a fundamental connection between two apparently unrelated mathematical ideas. This type of connection can be one of the most important aspects of a mathematical theorem, possibly surpassing the demonstration of the result in its depth.

Theorem 36. (*Archimedean Property of $\mathbb{R}$*) For all $x \in \mathbb{R}$ there exists $n \in \mathbb{N}$ such that $n > x$.

Definition 37. (*Absolute Value*) Let $x \in \mathbb{R}$. Then the absolute value of x , denoted $|x|$, is defined by

$$|x| := \begin{cases} x & x \geq 0 \\ -x & x < 0. \end{cases}$$

Theorem 38. $|xy| = |x| |y|$.

Theorem 39. $|x + y| \leq |x| + |y|$.

Theorem 40. $||x| - |y|| \leq |x - y|$.

Theorem 41. (*Denseness of $\mathbb{Q}$ in $\mathbb{R}$*) Suppose $x, y \in \mathbb{R}$ with $x < y$. Then there exists $r \in \mathbb{Q}$ such that $x < r < y$.

Problem 42. Show that the equation $x^2 = 2$ has a solution in $\mathbb{R}$. Note - this may not be easy. Work on it, and if you feel like your progress is beginning to slow, remember to revisit it later.

Problem 43. Just to be certain that there exist real numbers which are not rational, show that $\sqrt{2} \in \mathbb{I} := \mathbb{Q}^c$, the irrational numbers.

Problem 44. If $r \in \mathbb{Q}$ and $z \in \mathbb{I}$, then $r + z \in \mathbb{I}$.

Problem 45. (Between any two rational numbers there is an irrational number) Suppose $x, y \in \mathbb{Q}$ with $x < y$. Then there exists $z \in \mathbb{I}$ such that $x < z < y$.

Theorem 46. Let $S \subseteq \mathbb{R}$, be nonempty and bounded below. Prove $\inf S \in \mathbb{R}$.

Theorem 47. Let A and B be nonempty subsets of $\mathbb{R}$ such that $A \subseteq B$. Prove that if B is bounded above, then $\sup A \leq \sup B$, and if B is bounded below, $\inf B \leq \inf A$.

Theorem 48. Let A and B be nonempty subsets of $\mathbb{R}$ such that for all $x \in A$ and for all $y \in B$, one has $x \leq y$. Then $\sup A \leq \inf B$.

Theorem 49. Let $x, y \in \mathbb{R}$. Then $x = y$ if and only if for all $\varepsilon > 0$, $|x - y| < \varepsilon$.

Theorem 50. Let $x, y \in \mathbb{R}$. If for all $y' > y$ one has $x \leq y'$, then $x \leq y$.

Theorem 51. Let S be a nonempty set, bounded above, and let ℓ be an upper bound for S . Then $\ell = \sup S$ if and only if for all $\varepsilon > 0$, $S \cap (\ell - \varepsilon, \ell] \neq \emptyset$.

Problem 52. Formulate and prove a theorem analogous to Theorem 51 for $\ell = \inf S$.

Definition 53. (“Eventually”) Let $P(n)$ be a propositional function on $\mathbb{N}$ (a statement whose truth may depend on the value of the natural number n , like “ n is even”). We say $P(n)$ holds eventually if there exists $N \in \mathbb{N}$ such that $P(n)$ is true for all $n > N$.

Definition 54. A function f from $X \subseteq \mathbb{R}$ to $Y \subseteq \mathbb{R}$ is a set of ordered pairs (x, y) in the Cartesian product $X \times Y$, for which each $x \in X$ appears in exactly one ordered pair (x, y) . We write $f : X \rightarrow Y$ and say that f maps X to Y . We will adopt the standard functional notation $f(x)$ to represent the value of y in $(x, y) \in f$. The domain of f (denoted $\text{dom}(f)$) is X and the range of f (denoted $\text{ran}(f)$) is the set $\{y \in Y \mid \text{there exists } x \in X \text{ such that } y = f(x)\}$.

Definition 55. We say a function $f : X \rightarrow Y$ maps X into itself if $Y \subseteq X$.

Definition 56. Consider $s, t \in \mathbb{R}$ with $s < t$. An interval I is a subset of the real numbers of one of the following forms:

$$[s, t] := \{x \in \mathbb{R} \mid s \leq x \leq t\} \quad (1.2.1)$$

$$[s, t) := \{x \in \mathbb{R} \mid s \leq x < t\} \quad (1.2.2)$$

$$(s, t] := \{x \in \mathbb{R} \mid s < x \leq t\} \quad (1.2.3)$$

$$(s, t) := \{x \in \mathbb{R} \mid s < x < t\} \quad (1.2.4)$$

$$[s, \infty) := \{x \in \mathbb{R} \mid s \leq x\} \quad (1.2.5)$$

$$(s, \infty) := \{x \in \mathbb{R} \mid s < x\} \quad (1.2.6)$$

$$(-\infty, s] := \{x \in \mathbb{R} \mid x \leq s\} \quad (1.2.7)$$

$$(-\infty, s) := \{x \in \mathbb{R} \mid x < s\}. \quad (1.2.8)$$

An interval of the form (1.2.1) is called a closed interval, while an interval of the form (1.2.4) is called an open interval. The intervals of the form (1.2.1), (1.2.2), (1.2.3), and (1.2.4) are bounded intervals, and the others are unbounded intervals. From here on, when we write any of the first four interval types above, it is assumed that $s, t \in \mathbb{R}$ with $s < t$. It should be noted that, to maintain consistency with common practice, the particular letters used to represent the endpoints may vary depending on which section you're studying. Please be aware of this.

Chapter 2

Sequences of Real Numbers

Remark 57. Unless otherwise indicated, from here on the letters i, j, k, m , and n are assumed to represent natural numbers.

Definition 58. (Sequence) A sequence a is a function $a : \mathbb{N} \rightarrow \mathbb{R}$, whose ordered pairs are $(n, a(n))$. We will abuse the concept somewhat and permit ourselves to say that the sequence is the object comprising the ordered list of the second components (by convention denoted a_n instead of $a(n)$) of the sequence, with the order inherited from $\mathbb{N}$. That is, a sequence is an ordered infinite-tuple of real numbers, denoted $a = (a_n)_{n=1}^{\infty} = (a_1, a_2, a_3, \dots)$, or simply (a_n) . The a_n 's are called the terms of a .

Definition 59. The range of a sequence a , denoted $\text{ran}(a)$, is the set of all terms of a . If $\text{ran}(a) \subseteq X \subseteq \mathbb{R}$, then we say “ a is in X .” If $\text{ran}(a)$ is bounded above (below), we say “ a is bounded above (below)”. If a is bounded both above and below, we say “ a is bounded.”

Problem 60. Find the range of each of the following sequences:

1. $\left(\frac{1}{n}\right)$
2. $\left(\frac{(-1)^n}{n^2}\right)$
3. $\left(\frac{\cos n\pi}{4}\right)$
4. $\left(\frac{(3+(-1)^n)^n}{3^n}\right)$

In the next five definitions, let a represent a sequence.

Definition 61. (Convergence) We say that a converges when there exists $\ell \in \mathbb{R}$ such that for all $\varepsilon > 0$ there exists j such that for all $n > j$, $|a_n - \ell| < \varepsilon$. In this case, we'll call ℓ the limit of the sequence a . We write $a \rightarrow \ell$, $\lim_{n \rightarrow \infty} a_n = \ell$ or (in a mild abuse of notation) $a_n \rightarrow \ell$, and say “ a converges to ℓ ” or “ a has limit ℓ .” This is equivalent to the statement that for all $\varepsilon > 0$, eventually $|a_n - \ell| < \varepsilon$.

Problem 62. (Divergence) Formulate precisely the negation of the conditions in Definition 61; when the negated conditions are satisfied we say that the sequence a diverges. En route, precisely formulate the notions of $\lim_{n \rightarrow \infty} a_n = +\infty$ and $\lim_{n \rightarrow \infty} a_n = -\infty$. We will not use the terms “converges” or “converges to” in these circumstances.

Definition 63. (Subsequence) Let a be a sequence, let $k_0 \in \mathbb{N}$, and let $(n_k)_{k=k_0}^\infty$ be a strictly increasing sequence with $\text{ran}((n_k)) \subseteq \mathbb{N}$. Then the sequence (a_{n_k}) is called a subsequence of a .

Definition 64. (Tail of a Sequence) Let $a = (a_n)_{n=1}^\infty$ be a sequence. A tail of a is a subsequence of a of the form $(a_{n+k}) := (a_n)_{n=k}^\infty$ for some $k \in \mathbb{N}$.

Definition 65. (Monotonicity) Let m, n denote elements of $\mathbb{N}$. Then

$$\text{if for all } m > n, \begin{cases} a_m \geq a_n, & \text{then } a \text{ is increasing (nondecreasing)} \\ a_m > a_n, & \text{then } a \text{ is strictly increasing} \\ a_m \leq a_n, & \text{then } a \text{ is decreasing (nonincreasing)} \\ a_m < a_n, & \text{then } a \text{ is strictly decreasing.} \end{cases}$$

In each case we say that a is monotonic.

Definition 66. Let a, b be sequences and $\lambda \in \mathbb{R}$. Define the following sequences:

$$\begin{aligned} a \pm b &:= (a_1 \pm b_1, a_2 \pm b_2, \dots) \\ ab &:= (a_1 b_1, a_2 b_2, \dots) \\ \lambda a &:= (\lambda a_1, \lambda a_2, \dots) \\ \frac{a}{b} &:= \left(\frac{a_1}{b_1}, \frac{a_2}{b_2}, \dots \right) \quad \text{if } b_n \text{ never equals } 0 \end{aligned}$$

Theorem 67. Let a be a sequence. If $a \rightarrow m$ and $a \rightarrow \ell$, then $\ell = m$.

Theorem 68. Let a be a sequence. Then the following are equivalent:

1. $a \rightarrow \ell$
2. for every $k \in \mathbb{N}$, the tail (a_{n+k}) of a converges to ℓ , and
3. there exists $k \in \mathbb{N}$ such that the tail (a_{n+k}) of a converges to ℓ .

Theorem 69. If a is an eventually increasing sequence bounded above, then $a \rightarrow \sup \text{ran}(a)$.

Theorem 70. If a is an eventually decreasing sequence bounded below, then $a \rightarrow \inf \text{ran}(a)$.

Problem 71. Let $a_1 = 1$ and $a_{n+1} = \frac{1}{3}(a_n + 1)$ for all $n \in \mathbb{N}$. Show that $\ell := \lim_{n \rightarrow \infty} a_n$ exists, and find ℓ . Hint: Use induction.

Theorem 72. Let a be a sequence, and suppose a converges. Then $\text{ran}(a)$ is bounded.

Problem 73. The harmonic sequence $\left(\frac{1}{n}\right)$ converges.

Theorem 74. Suppose a is a sequence converging to $\ell \in \mathbb{R}$, and let $x, y \in \mathbb{R}$ such that $x < y$ and for all n , $a_n \in [x, y]$. Then $\ell \in [x, y]$.

Theorem 75. Suppose a and b are sequences, $\ell, m \in \mathbb{R}$ with $a \rightarrow \ell$ and $b \rightarrow m$. Then $a + b \rightarrow \ell + m$.

Theorem 76. Suppose a and b are sequences, $\ell, m \in \mathbb{R}$ with $a \rightarrow \ell$ and $b \rightarrow m$. Then $ab \rightarrow \ell m$.

Theorem 77. Suppose a is a sequence converging to $\ell \in \mathbb{R}$ and $\lambda \in \mathbb{R}$. Then $\lambda a \rightarrow \lambda \ell$.

Theorem 78. Suppose a, b are sequences which converge to ℓ, m respectively, and that for all n , $b_n \neq 0$ and also $m \neq 0$. Then $\frac{a}{b} \rightarrow \frac{\ell}{m}$.

Theorem 79. (Squeeze Theorem) Let a, b, c be sequences with $a \rightarrow \ell$ and $b \rightarrow \ell$, and suppose that eventually $a_n \leq c_n \leq b_n$. Then $c \rightarrow \ell$.

Theorem 80. Let a be a sequence with $a_n \rightarrow 0$ and b be a bounded sequence. Then $(a_n b_n)$ converges to 0.

Theorem 81. Let a be a sequence which converges to $\ell \in \mathbb{R}$. Then every subsequence of a converges to ℓ .

Definition 82. Let a be a bounded sequence. Define sequences (u_N) and (v_N) by, for all $N \in \mathbb{N}$,

$$u_N := \inf \{a_n \mid n > N\} \quad (2.0.1)$$

$$v_N := \sup \{a_n \mid n > N\} \quad (2.0.2)$$

Theorem 83. If a is a bounded sequence, then $\lim_{N \rightarrow \infty} u_N$ and $\lim_{N \rightarrow \infty} v_N$ exist.

Definition 84. (limsup and liminf) Let a be a bounded sequence, and let (u_N) and (v_N) be as defined in Equations (2.0.1) and (2.0.2), respectively. Define the limit inferior of a and limit superior of a , denoted $\liminf a$ and $\limsup a$, respectively, as

$$\liminf a := \lim_{N \rightarrow \infty} u_N$$

$$\limsup a := \lim_{N \rightarrow \infty} v_N.$$

Problem 85. Formulate precisely the extension of the notions of $\liminf$ and $\limsup$ to unbounded sequences, so that $\limsup a = +\infty$, $\limsup a = -\infty$, $\liminf a = -\infty$, and $\liminf a = +\infty$ are consistent with the solution to Exercise 62.

Theorem 86. Let a be a bounded sequence. Then $\limsup a = \ell \in \mathbb{R}$ if and only if both of the following hold:

1. For all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n > N$, $a_n < \ell + \varepsilon$, and
2. For all $\varepsilon > 0$ and for all $N \in \mathbb{N}$ there exists $n > N$ such that $a_n > \ell - \varepsilon$.

Problem 87. Formulate and prove the theorem analogous to Theorem 86 for $\liminf a$.

Theorem 88. Let a be a sequence with $a \rightarrow \ell$. Then $\limsup a \leq \ell$.

Theorem 89. Let a be a sequence with $a \rightarrow \ell$. Then $\ell \leq \liminf a$.

Theorem 90. If a is a sequence with $a \rightarrow \ell$, then $\liminf a = \ell = \limsup a$.

Theorem 91. Let a and b be sequences such that eventually $a_n \leq b_n$. Then $\liminf a \leq \liminf b$ and $\limsup a \leq \limsup b$.

Theorem 92. If a is a sequence with $\liminf a = \limsup a$, then either a converges to their common value, or diverges to one of $\pm\infty$.

Definition 93. (Cauchy Sequences) Let a be a sequence. If for all $\varepsilon > 0$ there exists j such that for all $m, n > j$, $|a_n - a_m| < \varepsilon$, we say that a is a Cauchy sequence (or simply “ a is Cauchy”).

Theorem 94. Cauchy sequences are bounded.

Theorem 95. Convergent sequences are Cauchy sequences.

Theorem 96. Cauchy sequences converge.

Problem 97. Let $\lambda > 0$, suppose $r \in (0, 1)$, and let a be a sequence such that for all n , $|a_{n+1} - a_n| < \lambda r^n$. Show that a is convergent.

Definition 98. Let a be a sequence. We'll say that the n^{th} term a_n of a is dominant if for all $m > n$, $a_m < a_n$.

Theorem 99. Every sequence has a monotonic subsequence. (Hint: Consider two cases, one that there are infinitely many dominant terms, the other that there are only finitely many dominant terms.)

Theorem 100. Let a be a bounded sequence. Then there exists a monotonic subsequence (a_{n_k}) converging to $\limsup a$ and a monotonic subsequence (a_{m_k}) converging to $\liminf a$. (Hint: Consider the same cases as in the hint for Theorem 99.)

Theorem 101. (Bolzano-Weierstrass) Every bounded sequence has a convergent subsequence.

Problem 102. Prove carefully that $\lim_{n \rightarrow \infty} \frac{3n^5 - 4n^3 + n - 1}{13n^5 - 200n^4 + n^2 + 12} = \frac{3}{13}$.

Theorem 103. $n^{\frac{1}{n}} \rightarrow 1$. Hint: Define $s_n := n^{\frac{1}{n}} - 1$ and use the Binomial Theorem

$$(x+y)^n = x^n + \binom{n}{1} x^{n-1}y + \cdots + \binom{n}{n-1} xy^{n-1} + y^n$$

Theorem 104. If $t > 0$, then $t^{\frac{1}{n}} \rightarrow 1$.

Theorem 105. If a is a sequence for which there exists j such that for all $n > j$, $a_n \neq 0$, then

$$\liminf \left| \frac{a_{n+1}}{a_n} \right| \leq \liminf |a_n|^{1/n} \leq \limsup |a_n|^{1/n} \leq \limsup \left| \frac{a_{n+1}}{a_n} \right|.$$

Chapter 3

Series of Real Numbers

Definition 106. An infinite series is the formal “sum” of a sequence; that is, if a is a sequence, then the series associated with a is $\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \cdots$. For convenience, some series may possess a different initial index such as $\sum_{n=0}^{\infty} a_n$. We will write $\sum a_n$ for the series $\sum_{n=1}^{\infty} a_n$.

Definition 107. Let $\sum a_n$ be a series. The sequence of partial sums of $\sum a_n$ is the sequence s defined by

$$s_n = \sum_{k=1}^n a_k = a_1 + a_2 + \cdots + a_n$$

for each $n \in \mathbb{N}$.

Definition 108. We say the series $\sum a_n$ converges if there exists $\ell \in \mathbb{R}$ such that its associated s converges to ℓ , and diverges if s diverges.

Remark 109. We will adopt the convention that for a convergent series $\sum a_n$ the notation $\sum a_n$ can be taken to mean the sequence of partial sums of the series, or the value of the sum. For example, for the series $\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$ (see Problem 110) we can say either

$$\text{“} \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n \text{ converges”} \quad \text{or} \quad \text{“} \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n = 1 \text{.”}$$

Problem 110. Let $c \in \mathbb{R} \setminus \{0\}$. Find all values of r so that the series $\sum_{n=0}^{\infty} cr^n$ converges, and prove that your answer is correct.

Theorem 111. If $\sum a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.

Problem 112. Prove that $\sum \frac{1}{n}$ diverges. Hint: Consider partial sums of the form s_{2^n} .

Remark 113. Note that by Exercise 112, we have an example of a sequence a for which $|a_{n+1} - a_n| < \frac{1}{n}$ for all $n \in \mathbb{N}$, but which diverges. See Exercise 97.

Problem 114. Show that if the sequence $a \rightarrow 0$, then there exists a subsequence (a_{n_k}) for which $\sum a_{n_k}$ converges. Can you obtain the same conclusion with some weaker hypothesis?

Definition 115. We say that a series $\sum a_n$ satisfies the Cauchy criterion if for all $\varepsilon > 0$ there exists j such that $n \geq m > j$ implies $\left| \sum_{k=m}^n a_k \right| < \varepsilon$.

Theorem 116. A series converges if and only if it satisfies the Cauchy criterion.

Definition 117. If a is a sequence for which the associated sequence of partial sums s is increasing and unbounded, then $\sum a_n := +\infty$.

Definition 118. (Absolute convergence) A series $\sum a_n$ is absolutely convergent if $\sum |a_n|$ converges.

Theorem 119. Absolutely convergent series converge.

Theorem 120. (Comparison Test) Let $\sum a_n$ be a series of nonnegative terms.

1. If $\sum a_n$ converges and $|b_n| \leq a_n$ eventually, then $\sum b_n$ converges.
2. If $\sum a_n = +\infty$ and eventually $b_n \geq a_n$, then $\sum b_n = +\infty$.

Theorem 121. If a is a convergent series of positive terms and if $p > 1$, then $\sum a_n^p$ converges.

Theorem 122. Let a, b be sequences with positive terms. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lambda \in \mathbb{R} \setminus \{0\}$, then $\sum a_n$ and $\sum b_n$ either both converge or both diverge.

Problem 123. Generalize Theorem 122 to the cases $\lambda = 0$ and $\lambda = +\infty$.

Theorem 124. (Cauchy Condensation Test) Let a be a decreasing sequence with nonnegative terms. Then $\sum a_n$ converges if and only if $\sum_{n=0}^{\infty} 2^n a_{2^n}$ converges.

Theorem 125. The series $\sum \frac{1}{n^p}$ converges if and only if $p > 1$.

Theorem 126. (Root Test) Let a be a bounded sequence and let $\alpha := \limsup |a_n|^{\frac{1}{n}}$. Then $\sum a_n$ converges if $\alpha < 1$, diverges if $\alpha > 1$, and can either converge or diverge if $\alpha = 1$.

Theorem 127. (Ratio Test) Let a be a sequence. Then $\sum a_n$ converges if $\limsup \left| \frac{a_{n+1}}{a_n} \right| < 1$, diverges if $\liminf \left| \frac{a_{n+1}}{a_n} \right| > 1$, and can either converge or diverge if

$$\liminf \left| \frac{a_{n+1}}{a_n} \right| \leq 1 \leq \limsup \left| \frac{a_{n+1}}{a_n} \right|.$$

Problem 128. Let $a > 0$. Show that $\left(\frac{a^n}{n!}\right)$ converges to 0. (Hint: The techniques of this section permit a more elegant proof of this result than those of Chapter 2.)

Theorem 129. (Alternating Series Test) If $\sum (-1)^n a_n$ is a series with (a_n) decreasing and $a_n \rightarrow 0$, then $\sum (-1)^n a_n$ converges.

Problem 130. Give an example of a divergent series $\sum a_n$ for which $\sum a_n^2$ converges, and an example of a convergent series $\sum b_n$ for which $\sum b_n^2$ diverges. Justify your assertions.

Definition 131. (Conditional convergence) A series $\sum a_n$ is termed conditionally convergent if $\sum a_n$ converges but $\sum |a_n|$ diverges.

Problem 132. Show that the Alternating Harmonic Series $\sum \frac{(-1)^n}{n}$ is conditionally convergent.

Definition 133. Let a be a sequence. A rearrangement $(a_{\sigma(n)})$ of (a_n) is the composition of the sequence with a bijection σ on $\mathbb{N}$ (technically this could be written as a composition $a \circ \sigma$ - see Definition 143 in Chapter 4); this is just a reordering of the terms of a . The series $\sum a_{\sigma(n)}$ associated with $(a_{\sigma(n)})$ is called a rearrangement of the series $\sum a_n$.

Problem 134. Let $a \rightarrow \ell$. Show that any rearrangement $(a_{\sigma(n)})$ of a converges to ℓ .

Problem 135. Fix $\lambda \in \mathbb{R}$. Show rigorously that there exists a rearrangement of the alternating harmonic series $\sum \frac{(-1)^n}{n}$ so that the rearranged series converges to λ .

Theorem 136. Let $\sum a_n$ be an absolutely convergent series converging to ℓ . Then if $\sum a_{\sigma(n)}$ is a rearrangement of $\sum a_n$, then $\sum a_{\sigma(n)}$ converges to ℓ .

Chapter 4

Functions

In this and each of the remaining sections of these notes f and g will refer to functions. In keeping with common practice, our notation for a sequence in this section will transition from a to (x_n) , and a and b may represent real numbers. The letters X, Y , and Z will represent subsets of $\mathbb{R}$.

Definition 137. (Continuity) A function $f : X \rightarrow Y$ is continuous at $x_0 \in X$ if for all sequences (x_n) in X converging to x_0 , $\lim_{n \rightarrow \infty} f(x_n) = f(x_0)$ holds. If f is continuous at every point in $S \subseteq \text{dom}(f)$, we say that f is continuous on S . If f is continuous at every point in $\text{dom}(f)$, then we say f is continuous.

Remark 138. The elementary trigonometric, exponential, and logarithmic functions and their respective properties are assumed; we'll permit ourselves to use the facts that $\sin x$, $\cos x$, and (if $a > 0$) a^x and $\log_a(x)$ are continuous.

Definition 139. Let f, g be functions and $\lambda \in \mathbb{R}$. The functions $f \pm g$, fg , f/g and λf are given by $(f \pm g)(x) = f(x) \pm g(x)$, $(fg)(x) = f(x)g(x)$, $(f/g)(x) = f(x)/g(x)$ (provided $g(x) \neq 0$), and $(\lambda f)(x) = \lambda \cdot f(x)$, respectively.

Theorem 140. Let $\lambda \in \mathbb{R}$. If $f : X \rightarrow Y$ is continuous at $x_0 \in X$, then the function λf is continuous at x_0 .

Theorem 141. If $f : X \rightarrow Y$ is continuous at $x_0 \in X$, then the function $|f|$ is continuous at x_0 .

Theorem 142. If $f : X \rightarrow Y$ is continuous at $x_0 \in X$, and if $g : X \rightarrow Y$ is continuous at $x_0 \in X$, then

$f + g$	is continuous at x_0
fg	is continuous at x_0
f/g	is continuous at x_0 if $g(x_0) \neq 0$

Definition 143. Suppose $g : X \rightarrow Y$ and $f : Y \rightarrow Z$. Then the composition $f \circ g : X \rightarrow Z$ of f with g is $(f \circ g)(x) := f(g(x))$ for each $x \in X$.

Theorem 144. Suppose $g : X \rightarrow Y$ and $f : Y \rightarrow Z$. If g is continuous at x_0 and f is continuous at $g(x_0)$, then $f \circ g$ is continuous at x_0 .

Definition 145. Let f and g be functions defined on a common domain. Define the functions $\max\{f, g\}$ and $\min\{f, g\}$ by, for each x in that domain,

$$\begin{aligned}\max\{f, g\}(x) &:= \max\{f(x), g(x)\} \\ \min\{f, g\}(x) &:= \min\{f(x), g(x)\}.\end{aligned}$$

Problem 146. Find functions h_1 and h_2 which yield $\max\{f, g\} = \frac{1}{2}(f + g) + h_1$ and $\min\{f, g\} = \frac{1}{2}(f + g) + h_2$.

Theorem 147. If f and g are continuous at x_0 , then both $\max\{f, g\}$ and $\min\{f, g\}$ are continuous at x_0 .

Problem 148. Show that $f(x) = \sqrt{x}$ is continuous on $[0, \infty)$.

Problem 149. Show that $f(x) = \begin{cases} x^3 & x \in \mathbb{Q} \\ 0 & x \in \mathbb{I} \end{cases}$ is continuous at x_0 if and only if $x_0 = 0$.

Problem 150. Fix $m \in \mathbb{N}$. Prove that the polynomial $f : \mathbb{R} \rightarrow \mathbb{R}$ with

$$f(x) = a_m x^m + a_{m-1} x^{m-1} + \cdots + a_1 x + a_0, \quad a_m, a_{m-1}, \dots, a_0 \in \mathbb{R}$$

is continuous.

Theorem 151. $f : X \rightarrow Y$ is continuous at $x_0 \in X$ if and only if for all $\varepsilon > 0$ there exists $\delta > 0$ such that $x \in X$ with $|x - x_0| < \delta$ implies $|f(x) - f(x_0)| < \varepsilon$.

Problem 152. Can we replace either of the latter two strict inequalities in Theorem 151 with non-strict inequalities?

Problem 153. Prove that the polynomial given by $f(x) = 9x^2 + 4$ is continuous by using Theorem 151.

Definition 154. Let $f : X \rightarrow Y$ and suppose $X \cap Y \neq \emptyset$. A fixed point of f is $x \in X$ such that $f(x) = x$.

Definition 155. (Contraction) A function f mapping $[a, b]$ into itself is a contraction if there exists $r \in (0, 1)$ such that for all $x, y \in [a, b]$, we have $|f(x) - f(y)| \leq r|x - y|$.

Theorem 156. If f is a contraction then f is continuous.

Theorem 157. If f is a contraction then f has a unique fixed point.

Theorem 158. Let f be a contraction. If z is in the domain of f , then the sequence c defined by $c_1 = z$, $c_{n+1} = f(c_n)$ converges to the fixed point of f .

Remark 159. The process of feeding the output of a function back into the function, as described in Theorem 158, is denoted by the phrase “repeated iteration.”

Problem 160. Let $w > 0$, and let f be defined by $f(x) = \frac{1}{2}\left(x + \frac{w}{x}\right)$ with $\text{dom}(f) = \left[\sqrt{\frac{w}{2}}, 2\sqrt{\frac{w}{2}}\right]$. Show that f is a contraction.

Problem 161. Show that repeated iteration of the function described in Exercise 160, starting with a point in the domain of f , results in a sequence which converges to $\sqrt{w}$.

Problem 162. Prove that $f(x) = \begin{cases} \frac{1}{x} \sin\left(\frac{1}{x^3}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$ is not continuous at $x = 0$.

Theorem 163. Let f be continuous on a closed interval. Then f is bounded.

Theorem 164. Let f be continuous on a closed interval I . Then the set $\{f(x) \mid x \in I\}$ has a maximum and a minimum; that is, there exist $x_0, y_0 \in I$ such that $f(x_0) = \max_{x \in I} f(x)$ and $f(y_0) = \min_{x \in I} f(x)$. In other words, f assumes its minimum and maximum values. Hint: Construct a sequence (x_n) for which $f(x_n)$ converges to the supremum of $\{f(x) \mid x \in I\}$, apply the Bolzano-Weierstrass Theorem, and invoke the continuity of f .

Theorem 165. (Intermediate Value Theorem) If f is continuous on an interval I , and if $x, y \in I$ such that $x < y$, $f(x) \neq f(y)$, and $t \in \mathbb{R}$ with either $f(x) < t < f(y)$ or $f(y) < t < f(x)$, then there exists $s \in (x, y)$ such that $f(s) = t$.

Definition 166. Let $S \subseteq \mathbb{R}$, and $f : S \rightarrow \mathbb{R}$. Then f is uniformly continuous on S if for all $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, y \in S$ with $|x - y| < \delta$ one has $|f(x) - f(y)| < \varepsilon$.

Theorem 167. If f is continuous on a $[a, b]$ then f is uniformly continuous on $[a, b]$.

Problem 168. Show that $f(x) = \sin\left(\frac{1}{x^2}\right)$ is uniformly continuous on (a, b) for any $a, b > 0$.

Theorem 169. If f is uniformly continuous on $S \subseteq \mathbb{R}$ and (x_n) is a Cauchy sequence with $\text{ran}((x_n)) \subseteq S$, then $(f(x_n))$ is a Cauchy sequence.

Problem 170. Let $S \subseteq \mathbb{R}$, and $f : S \rightarrow \mathbb{R}$. Then f is not uniformly continuous on S if and only if there exists $\varepsilon > 0$ and sequences (x_n) and (y_n) in S such that for all $n \in \mathbb{N}$, $|x_n - y_n| < \frac{1}{n}$, but $|f(x_n) - f(y_n)| \geq \varepsilon$.

Problem 171. Show that $f(x) = \cos\left(\frac{1}{x^3}\right)$ is not uniformly continuous on $(0, b]$ for any $b > 0$.

Chapter 5

Sequences and Series of Functions

Remark 172. Notational worlds collide; back to $a = (a_n)$ for real sequences.

Definition 173. Let a be a sequence. The series $\sum_{n=0}^{\infty} a_n x^n$, where x is a real variable, is called a power series.

Definition 174. Let a be a sequence, and let α be as in Theorem 126. If $\alpha = 0$ define $R_a := +\infty$; if $\alpha = +\infty$ define $R_a = 0$; otherwise define $R_a := \frac{1}{\alpha}$.

Theorem 175. $\sum_{n=0}^{\infty} a_n x^n$ converges if $|x| < R_a$ and diverges if $|x| > R_a$.

Definition 176. (Radius of Convergence) The nonnegative real number R_a defined in Definition 174 and referred to in Theorem 175 is called the radius of convergence of $\sum_{n=0}^{\infty} a_n x^n$. When no ambiguity is possible, we'll use R instead of R_a .

Theorem 177. The set of points $I_a \subseteq \mathbb{R}$ for which $\sum_{n=0}^{\infty} a_n x^n$ converges is one of the following: if $R_a = +\infty$, $I_a = \mathbb{R}$; if $R_a = 0$, $I_a = \{0\}$; otherwise I_a is one of $(-R_a, R_a)$, $[-R_a, R_a]$, $(-R_a, R_a]$, or $[-R_a, R_a)$.

Definition 178. (Interval of Convergence) The set I_a referred to in Theorem 177 for which $\sum_{n=0}^{\infty} a_n x^n$ converges is called the interval of convergence of $\sum_{n=0}^{\infty} a_n x^n$. When no ambiguity is possible, we'll use I instead of I_a .

Problem 179. Find sequences a, b, c, d, e, f such that

$$\begin{aligned} \sum_{n=0}^{\infty} a_n x^n &\text{converges on all of } \mathbb{R}, \\ \sum_{n=0}^{\infty} b_n x^n &\text{converges at and only at } x = 0, \\ \sum_{n=0}^{\infty} c_n x^n &\text{converges on and only on } (-R_c, R_c), \end{aligned}$$

$$\sum_{n=0}^{\infty} d_n x^n \text{ converges on and only on } [-R_d, R_d],$$

$$\sum_{n=0}^{\infty} e_n x^n \text{ converges on and only on } (-R_e, R_e], \text{ and}$$

$$\sum_{n=0}^{\infty} f_n x^n \text{ converges on and only on } [-R_f, R_f).$$

Problem 180. Find I for $\sum_{n=0}^{\infty} 3^{-n} x^{5n}$.

Definition 181. (Pointwise convergence) Let f_n be a sequence of functions each of which is defined on $D \subseteq \mathbb{R}$. If there exists f on D such that for all $x \in D$, $f_n(x) \rightarrow f(x)$, then we say that (f_n) converges pointwise to f , call f the pointwise limit of (f_n) , and write $f_n \xrightarrow{p} f$.

Problem 182. Show that the pointwise limit of a sequence of continuous functions is not necessarily continuous.

Definition 183. (Uniform convergence) Let (f_n) be a sequence of functions defined on $D \subseteq \mathbb{R}$. Then (f_n) converges uniformly to f on D if for all $\varepsilon > 0$ there exists j such that for all $x \in D$ and for all $n > j$, $|f_n(x) - f(x)| < \varepsilon$; in this case we write $f_n \xrightarrow{u} f$.

Problem 184. Prove that $\left(\frac{x}{n}\right)$ converges uniformly to the zero function on any bounded interval but does not converge uniformly on $\mathbb{R}$.

Theorem 185. If (f_n) is a sequence of continuous functions on $D \subseteq \mathbb{R}$ converging uniformly to a function f , then f is continuous on D .

Problem 186. Show that a power series is uniformly convergent on $\mathbb{R}$ if and only if it is a polynomial.

You may receive credit for at most one of the next four problems.

Problem 187. For $n \in \mathbb{N}$, let $f_n(x) = \frac{x^n}{1+x^n}$ on $[0, \infty)$. Find the pointwise limit function, and determine where the convergence is uniform.

Problem 188. For $n \in \mathbb{N}$, let $f_n(x) = \frac{nx}{1+n^2x^2}$ on $[0, \infty)$. Find the pointwise limit function, and determine where the convergence is uniform.

Problem 189. For $n \in \mathbb{N}$, let $f_n(x) = \frac{nx}{1+nx}$ on $[0, \infty)$. Find the pointwise limit function, and determine where the convergence is uniform.

Problem 190. For $n \in \mathbb{N}$, let $f_n(x) = \frac{1}{1+x^n}$ on $[0, \infty)$. Find the pointwise limit function, and determine where the convergence is uniform.

Problem 191. Let (f_n) be a sequence of continuous functions on $[a, b]$ converging uniformly to f on $[a, b]$. Show that if (x_n) is a convergent sequence with range in $[a, b]$ and whose limit $x \in [a, b]$, then $f_n(x_n) \rightarrow f(x)$.

Theorem 192. Let $\sum_{n=0}^{\infty} f_n$ be a uniformly convergent series of continuous functions on $X \subseteq \mathbb{R}$. Then $\sum_{n=0}^{\infty} f_n$ is continuous.

Definition 193. (Uniformly Cauchy) Let (f_n) be a sequence of functions on $X \subseteq \mathbb{R}$. Then if for all $\varepsilon > 0$ there exists j such that for all $x \in X$ and for all $m, n > j$, $|f_n(x) - f_m(x)| < \varepsilon$, we say that (f_n) is uniformly Cauchy on X .

Theorem 194. Let (f_n) be a sequence of functions which is uniformly Cauchy on $X \subseteq \mathbb{R}$. Then there exists a function f on X with $f_n \xrightarrow{u} f$. Hint: First prove that the pointwise limit function exists, and use Definition 193 and Theorem 74.

Definition 195. We say that a series of functions $\sum_{k=0}^{\infty} f_k$ satisfies the Cauchy criterion uniformly on $X \subseteq \mathbb{R}$ if for all $\varepsilon > 0$ there exists j such that for all $x \in X$,

$$n \geq m > j \quad \Rightarrow \quad \left| \sum_{k=m}^n f_k(x) \right| < \varepsilon.$$

(Compare to Definition 115.)

Theorem 196. If $\sum_{k=0}^{\infty} f_k$ satisfies the Cauchy criterion uniformly on $X \subseteq \mathbb{R}$, then the series converges uniformly on X .

Theorem 197. (Weierstrass M-Test) Let (M_k) be a sequence of nonnegative real numbers with $\sum_{k=0}^{\infty} M_k < \infty$, and suppose that for all $k \in \mathbb{N}$, $|f_k(x)| \leq M_k$ for each $x \in X$. Then $\sum_{k=0}^{\infty} f_k$ converges uniformly on X .

Theorem 198. Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series with (possibly infinite) radius of convergence $R > 0$. If $b \in (0, R)$, then $\sum_{n=0}^{\infty} a_n x^n$ converges uniformly on $[-b, b]$ to a continuous function.

Chapter 6

Differentiation

Definition 199. (Differentiability) Suppose f is a function defined on an interval I . We say that f is differentiable at $x_0 \in I$ if

$$\lim_{n \rightarrow \infty} \frac{f(x_n) - f(x_0)}{x_n - x_0}$$

exists, is finite and equal for all sequences (x_n) for which $\text{ran}((x_n)) \subseteq I \setminus \{x_0\}$, and which converge to x_0 . If this is the case we write

$$f'(x_0) := \lim_{n \rightarrow \infty} \frac{f(x_n) - f(x_0)}{x_n - x_0}.$$

If f is differentiable everywhere on its domain, we say that f is differentiable.

Remark 200. Note that by Theorem 67, when it exists the number $f'(x_0)$ is unique.

Problem 201. Reformulate Definition 199 as an ε - δ - type statement.

Theorem 202. If a function f is differentiable at $x_0 \in \mathbb{R}$, then f is continuous at x_0 .

Problem 203. Give an example to show that the converse of Theorem 202 is false; justify your assertion.

Theorem 204. Let I be an interval in $\text{dom}(f)$, and $x_0 \in I$. If $c \in \mathbb{R}$ and f is differentiable at x_0 , then the function cf is differentiable at x_0 and $(cf)'(x_0) = cf'(x_0)$.

Remark 205. For the next three theorems, let I be an interval in $\text{dom}(f) \cap \text{dom}(g)$, and $x_0 \in I$.

Theorem 206. If f and g are both differentiable at x_0 , then the function $f + g$ is differentiable at x_0 and $(f + g)'(x_0) = f'(x_0) + g'(x_0)$.

Theorem 207. If f and g are differentiable at x_0 , then fg is differentiable at x_0 , and $(fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0)$.

Theorem 208. Let f and g be differentiable at x_0 , and suppose $g(x_0) \neq 0$. Then $\frac{f}{g}$ is differentiable at x_0 , and $\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{[g(x_0)]^2}$.

Problem 209. Let $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$ and $g(x) = x$ for $x \in \mathbb{R}$. Prove that f

and g are differentiable on $\mathbb{R}$. Find a sequence (x_n) for which $\lim_{n \rightarrow \infty} \frac{g(f(x_n)) - g(f(0))}{f(x_n) - f(0)}$ does not exist.

Theorem 210. Let f and g be functions with $\text{ran}(g) \subseteq \text{dom}(f)$. If g is differentiable at x_0 and f is differentiable at $g(x_0)$, then $f \circ g$ is differentiable at x_0 and $(f \circ g)'(x_0) = g'(x_0)f'(g(x_0))$.

Definition 211. (Local Extrema) Suppose f is a function on an open interval I . Then f has a local maximum (resp. local minimum) at $x_0 \in I$ if there exists $\delta > 0$ such that for all $y \in I \cap (x_0 - \delta, x_0 + \delta)$, $f(y) \leq f(x_0)$ (resp. $f(y) \geq f(x_0)$).

Theorem 212. Suppose f is a function on an open interval I . If f has a local maximum or a local minimum at $x_0 \in I$, and if f is differentiable at x_0 , then $f'(x_0) = 0$.

Theorem 213. (Rolle's Theorem) If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there exists $x_0 \in (a, b)$ such that $f'(x_0) = 0$.

Theorem 214. (Mean-Value Theorem) If f is continuous on $[a, b]$ and differentiable on (a, b) , then there exists $x_0 \in (a, b)$ such that $f'(x_0) = \frac{f(b) - f(a)}{b - a}$.

Problem 215. Determine if the hypotheses of Theorem 214 (or the first two hypotheses of Theorem 213) are equivalent to “ f is differentiable on $[a, b]$.”

Theorem 216. If f is differentiable on $[a, b]$ with $f'(x) = 0$ for all $x \in [a, b]$, then f is constant on $[a, b]$.

Problem 217. Define what it means for a function to be each of the four senses of monotonic on $[a, b]$ (see Definition 65), and state and prove what each of those cases imply for $f'(x)$ on $[a, b]$.

Theorem 218. Let (f_n) be a sequence of differentiable functions on $[a, b]$. If for all $x \in [a, b]$ $f_n(x) \rightarrow f(x)$, and $f'_n \xrightarrow{u} g$, then f is differentiable and $f'(x) = g(x)$ for all $x \in [a, b]$. Hint: Argue that if $y \in [a, b]$ then $f'(y)$ exists and equals $g(y)$, by

$$\begin{aligned} \left| \frac{f(x_n) - f(y)}{x_n - y} - g(y) \right| &\leq \left| \frac{f(x_n) - f(y)}{x_n - y} - \frac{f_n(x_n) - f_n(y)}{x_n - y} \right| \\ &\quad + \left| \frac{f_n(x_n) - f_n(y)}{x_n - y} - f'_n(y) \right| + |f'_n(y) - g(y)| \end{aligned}$$

where the third term can be controlled by hypothesis, the second by differentiability of f_n , and the first (more difficult) by appealing to the Mean Value Theorem and the idea of “uniformly Cauchy”, and using the fact that if a_n converges to ℓ , then $\lim_{n \rightarrow \infty} (a_m - a_n) = a_m - \ell$.

Theorem 219. Suppose (f_n) is a sequence of differentiable functions on $[a, b]$ and that $f'_n \xrightarrow{u} g$ on $[a, b]$. If there exists $y \in [a, b]$ such that $(f_n(y))$ is convergent, then there exists f on $[a, b]$ such that $f_n \xrightarrow{u} f$, f is differentiable and $f' = g$ on $[a, b]$. Hint:

$$|f_n(x) - f_m(x)| \leq |(f_n(x) - f_m(x)) - (f_n(y) - f_m(y))| + |(f_n(y) - f_m(y))|$$

and use the MVT.

Theorem 220. Suppose (f_n) is a sequence of differentiable functions on $[a, b]$ and that $\sum f'_n(x)$ converges uniformly to $g(x)$ on $[a, b]$. Then if there exists $y \in [a, b]$ such that $\sum f_n(y)$ converges, then $\sum f_n$ converges uniformly to a differentiable function f satisfying $f' = g$ on $[a, b]$.

Remark 221. Theorem 220 gives sufficient conditions for when $f(x) = \sum f_n(x)$ implies $f'(x) = \sum f'_n(x)$.

Theorem 222. Suppose $\sum_{n=0}^{\infty} a_n x^n$ has radius of convergence $R > 0$. Then $\sum_{n=1}^{\infty} n a_n x^{n-1}$ has radius of convergence R , and the convergence is uniform on closed intervals contained in $(-R, R)$.

Theorem 223. Suppose $f(x) := \sum_{n=0}^{\infty} a_n x^n$ has radius of convergence $R > 0$. Then f is differentiable on $(-R, R)$ and $f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$ for $x \in (-R, R)$.

Chapter 7

Integration

Remark 224. Throughout this section, unless otherwise indicated, we let $a, b \in \mathbb{R}$ with $a < b$, the symbol I will denote the closed interval $[a, b]$, and f will denote a bounded function on I .

Definition 225. (Partition, Mesh, Subinterval) A partition P of I is a set of at least three elements of $[a, b]$, containing a and b and at least one element in (a, b) . We write

$$P = \{a = t_0, t_1, \dots, t_{n-1}, t_n = b\}$$

and assume that $t_{k-1} < t_k$ for $k = 1, 2, \dots, n$. We'll denote the k^{th} subinterval $[t_{k-1}, t_k]$ as I_k and the length $t_k - t_{k-1}$ of I_k as ℓ_k . The mesh of P , denoted $\text{mesh}(P)$, equals $\max_k \{\ell_k\}$.

Definition 226. (Upper and Lower Darboux sums) Let f be a function defined on I , and P a partition. Define

$$\begin{aligned} m_k(f) &:= \inf \{f(x) \mid x \in I_k\} \\ M_k(f) &:= \sup \{f(x) \mid x \in I_k\}. \end{aligned}$$

Further define the upper and lower Darboux sums of f on I relative to P as

$$\begin{aligned} U_P(f) &:= \sum_{k=1}^n M_k(f) \ell_k \\ L_P(f) &:= \sum_{k=1}^n m_k(f) \ell_k \end{aligned}$$

Theorem 227. Let f be defined on I . Then for all partitions P of I , $L_P(f) \leq U_P(f)$.

Theorem 228. Let f be defined on I . Then for all partitions P and Q of I with $P \subseteq Q$,

$$L_P(f) \leq L_Q(f) \leq U_Q(f) \leq U_P(f).$$

Theorem 229. Let f be defined on I . Then for all partitions P and Q of I , $L_P(f) \leq U_Q(f)$.

Definition 230. (Darboux integrability) Let f be defined on I .

$$\begin{aligned} U(f) &:= \inf \{U_P(f) \mid P \text{ is a partition of } I\} \\ L(f) &:= \sup \{L_P(f) \mid P \text{ is a partition of } I\}. \end{aligned}$$

The quantities $U(f)$ and $L(f)$ are called the upper and lower Darboux integrals of f , respectively. If $L(f) = U(f)$, we say that f is Darboux integrable on $[a, b]$; in this case denote the value $\int_a^b f := U(f) (= L(f))$ as the Darboux integral of f on $[a, b]$.

Problem 231. Show that $f(x) = x$ is Darboux integrable on $[0, b]$ where $b > 0$.

Problem 232. Show that $f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \in \mathbb{I} \end{cases}$ is not Darboux integrable on any closed interval.

Theorem 233. Let f be defined on I . Then $L(f) \leq U(f)$.

Theorem 234. f is Darboux integrable on I if and only if for all $\varepsilon > 0$ there exists a partition P of $[a, b]$ such that $U_P(f) - L_P(f) < \varepsilon$.

Theorem 235. f is Darboux integrable on I if and only if for all $\varepsilon > 0$ there exists $\delta > 0$ such that $\text{mesh}(P) < \delta$ implies $U_P(f) - L_P(f) < \varepsilon$.

Theorem 236. If f is defined on I , and there exists a sequence (P_n) of partitions of I such that $\lim_{n \rightarrow \infty} (U_{P_n}(f) - L_{P_n}(f)) = 0$, then f is Darboux integrable on I , and $\int_a^b f = \lim_{n \rightarrow \infty} U_{P_n}(f) = \lim_{n \rightarrow \infty} L_{P_n}(f)$.

Definition 237. (Riemann integrability) A Riemann sum associated with f on I with respect to P is a sum of the form

$$R_n(f) := \sum_{k=1}^n f(x_k) \ell_k$$

where $x_k \in I_k$ for $k \in \{1, \dots, n\}$. We say that f is Riemann integrable on I if and only if there exists $R(f) \in \mathbb{R}$ such that for all $\varepsilon > 0$ there exists $\delta > 0$ such that $|R_n(f) - R(f)| < \varepsilon$ for every Riemann sum associated with f on I with respect to P where $\text{mesh}(P) < \delta$. If f is Riemann integrable on I , we denote $R(f)$ as the Riemann integral of f .

Theorem 238. f is Riemann integrable on I if and only if $R(f) = \int_a^b f$.

Definition 239. We say that if f is either Darboux or Riemann integrable on I , then f is integrable on I , and denote the common value of the Riemann and Darboux integrals as $\int_a^b f$.

Theorem 240. If f is integrable on $[a, b]$, then f is integrable on $[c, d]$ where $a < c < d < b$.

Theorem 241. If f is monotonic on I then f is integrable on I .

Theorem 242. If f is continuous on I then f is integrable on I .

Theorem 243. Let f and g be integrable on I . Then $f + g$ is integrable on I , and

$$\int_a^b f + g = \int_a^b f + \int_a^b g$$

Theorem 244. Let f be integrable on I , and let $c \in \mathbb{R}$. Then cf is integrable on I , and

$$\int_a^b cf = c \int_a^b f$$

Theorem 245. If f and g are both integrable on I and for all $x \in I$, $f(x) \leq g(x)$, then $\int_a^b f \leq \int_a^b g$.

Remark 246. Let f be a bounded function defined on $S \subseteq \mathbb{R}$. Denote $M(f, S) := \sup_{x \in S} \{f(x)\}$ and $m(f, S) := \inf_{x \in S} \{f(x)\}$.

Theorem 247. If S is any subset of I then $M(|f|, S) - m(|f|, S) \leq M(f, S) - m(f, S)$.

Theorem 248. If f is integrable on I , then $|f|$ is integrable on I and $\left| \int_a^b f \right| \leq \int_a^b |f|$.

Problem 249. Give an example of a function f on $[0, 1]$ for which f is not integrable but $|f|$ is integrable, and prove your assertion.

Theorem 250. Let f be both bounded and integrable on I . Show that f^2 is integrable on I .

Theorem 251. If f and g are integrable on I , then fg is integrable on I . Hint: Simplify $(f + g)^2 + (f - g)^2$.

Theorem 252. If f and g are both integrable on I , then $\max\{f, g\}$ and $\min\{f, g\}$ are integrable on I .

Theorem 253. If f is defined on I , $c \in (a, b)$ and if f is integrable on $[a, c]$ and on $[c, b]$, then f is integrable on I and

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

Theorem 254. Let $f_n \xrightarrow{u} f$ on $[a, b]$, and suppose each f_n is integrable. Then f is integrable and $\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n$.

Problem 255. Give an example of a sequence of functions (f_n) which converges pointwise to zero, but for which $\int_0^1 f_n = 1$ for every $n \in \mathbb{N}$. Comment on the applicability of Theorem 254.

Theorem 256. (Mean Value Theorem for Integrals) If f is continuous on I then there exists $x \in I$ such that

$$f(x) = \frac{1}{b-a} \int_a^b f.$$

Theorem 257. (Fundamental Theorem of Calculus I) Let f be continuous on I and differentiable on (a, b) . Then if f' is integrable on I , then

$$\int_a^b f' = f(b) - f(a).$$

Theorem 258. (Fundamental Theorem of Calculus II) Let f be integrable on I . Then if $x \in I$, define

$$F(x) := \int_a^x f(t) dt.$$

Then F is continuous on I . If f is continuous at $x_0 \in (a, b)$, then F is differentiable at x_0 and $F'(x_0) = f(x_0)$.

Theorem 259. If f and g are continuous functions on I and differentiable on (a, b) and if f' and g' are integrable on I , then

$$\int_a^b f'g + \int_a^b fg' = f(b)g(b) - f(a)g(a).$$

Theorem 260. Let g be a differentiable function on an open interval J on which g' is continuous, and I be an open interval such that for all $x \in J$, $g(x) \in I$. If f is continuous on I , then $f \circ g$ is continuous on J and

$$\int_a^b (f \circ g)g' = \int_{g(a)}^{g(b)} f$$

for $a, b \in J$.

Theorem 261. If f has finitely many discontinuities in I , then f is integrable on I .

Problem 262. Show that $f(x) = \begin{cases} 1 & x = \frac{1}{n} \text{ for some } n \in \mathbb{N} \\ 0 & \text{otherwise} \end{cases}$ is integrable on $[0, 1]$, and

find the value of $\int_0^1 f$.

You may receive credit for at most one of the next two problems.

Problem 263. Show that if f is continuous on $\mathbb{R}$ and for all $x \in \mathbb{R}$, $F(x) := \int_{2x-1}^{2x+1} f(t) dt$, then F is differentiable and compute F' .

Problem 264. Show that if f is continuous on $\mathbb{R}$ and for all $x \in \mathbb{R}$, $G(x) := \int_0^{\sin x} f(t) dt$, then G is differentiable and compute G' .

You may receive credit for at most one of the next two problems.

Problem 265. Find the value of $\int_0^{\sqrt{3}} \frac{x}{\sqrt{x^2+1}} dx$, justifying your answer.

Problem 266. Find the value of $\int_0^{\frac{\pi}{2}} x^2 \sin x dx$, justifying your answer.

Theorem 267. Suppose that $\sum f_n$ is a series of continuous functions f_n on I which converge uniformly to a function f on I . Then $\int_a^b f = \sum \int_a^b f_n$.